

JACK NGO

Denton, TX

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Education

University of North Texas

Aug 2024 – May 2028

B.S. Computer Science

Denton

- Maintained a 3.9/4.0 GPA while studying Computer Science at University of North Texas
- Completed coursework in Data Structures & Algorithms, System Programming, Internet Programming, Database Systems, Software Engineering, and Linear Algebra

Experience

Software Engineering Intern

Feb 2026 – May 2026

Eudaimonic Inc

Remote

- **Shipped 3+ full-stack React applications** with live **multi-user sync and presence**, **60fps** game loops, and drift-free sequencer timing on a 25ms scheduler tick, using **React, TypeScript, Web Audio API, and Express/MongoDB**.
- Cut storage writes by an **estimated 70%** by debouncing autosave, keeping transport state client-side, and computing progression state on read instead of persisting it, **using React, TypeScript, and a SQLite-backed record store**.
- **Rendered AI-generated slides with 100% compile success across a 5,000-slide benchmark**, by building a browser-side TSX-to-React compiler using **React, TypeScript, and Sucrase**.
- Shipped an **AI slide-generating agent** that generates and edits live React slides, by building a two-stage planner-builder pipeline using the **Vercel AI SDK, Zod, and Anthropic/OpenAI** models.
- **Prevented uncapped AI spend** on the company account while preserving no-signup trial access, by designing a **server-side, token-scoped billing architecture** using **Cloudflare Workers, Hono, and JWT auth**.

Research Assistant

Jul 2025 – Dec 2025

High Performance Cloud Computing Lab, UNT

University of North Texas

- Developed a **multi-threaded Piper TTS engine** with hierarchical message queues for real-time audio on SmartSight glasses
- Implemented a priority-queue interruption mechanism enabling urgent alerts to preempt passive speech
- Validated real-time audio systems end-to-end via automated test suites and keyboard-triggered threads

Student Researcher

May 2025 – Jul 2025

UNT AI Summer Research

University of North Texas

- Developed an **end-to-end Python UAV** imagery pipeline to **automate** Ligustrum sinense detection with geography researchers
- Achieved **F1 score up to 91%** classifying Ligustrum sinense by training **YOLO** and **Faster R-CNN** with a **ResNet-101 backbone**
- Authored unit tests for a **YOLOv8** annotation tool and auto-labeled **600+ images** to reduce manual labeling effort

Projects

Document Extraction Platform | *Node, Express, SQLite, LLM, sqlite-vec*

Jul 2026 – Present

- Architected a **multi-tenant platform** with per-tenant SQLite databases and a shared admin database for auth and job scheduling
- Developed an async pipeline utilizing SQLite change data capture on WAL inserts for Tesseract OCR and LLM zero-shot extraction
- Designed a hybrid query engine fusing SQL filters with semantic search via reciprocal rank fusion

Agentic RAG Support Bot | *Python, LangGraph, FastAPI, ChromaDB, BM25*

Jul 2026

- Processed 247 help-center pages into a bilingual RAG pipeline using hybrid retrieval and reranking
- Designed a self-correcting LangGraph agent executing retry logic up to 3x prior to escalating low-confidence answers to handoff
- Built a **39-question bilingual** eval harness scoring retrieval recall and RAGAS faithfulness to meet strict quality benchmarks

Skills

Languages: Python, TypeScript, JavaScript, C/C++ (OOP), SQL

Frameworks & Libraries: React, Node.js, Express, FastAPI (basic), LangGraph (basic), YOLO

Databases & Storage: SQLite, MongoDB (basic), ChromaDB (basic)

Tools & Technologies: Git, Linux, WebSocket(basic), Tesseract OCR

Soft Skills: Communication, Teamwork, Time Management, Organization, Proactive